

CLINICAL TRIALS AND OBSERVATIONS

Anti-BCMA/GPRC5D CAR T cells in patients with relapsed or refractory multiple myeloma who have extraosseous extramedullary disease

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KEY POINTS

- Anti-BCMA/GPRC5D CAR T cells yielded higher response rates and deeper responses in patients with RRMM with EMD.
- Anti-BCMA/GPRC5D CAR T cells showed a manageable safety profile, and the toxicities were comparable to single-target CAR T-cell products.

Patients with relapsed or refractory multiple myeloma (RRMM) with extraosseous extramedullary disease (EMD) have inferior outcomes and lack effective therapies. We developed anti-B-cell maturation antigen (anti-BCMA)/G protein-coupled receptor, class C group 5 member D (GPRC5D) bispecific chimeric antigen receptors (CARs) to investigate the activity and safety of the CAR T cells in patients with extraosseous EMD. In this single-arm, open-label, phase 2 trial, we enrolled 37 patients with RRMM with extraosseous EMD, and anti-BCMA/GPRC5D bispecific CAR T cells were administered at 2.0×10^6 CAR T cells per kg. At a median follow-up of 10.1 months (interquartile range, 6.4-19.1), 36 of 37 patients (97%) obtained an overall response and measurable residual disease negativity, including 16 (43%) with stringent complete response. The median progression-free survival was 5.8 months (95% confidence interval, 2.2-9.4), and the median overall survival was not reached. The most common grade 3 or worse adverse events were hematologic toxicities (except lymphopenia; 37/37). Twenty-seven patients (73%) experienced cytokine release syndrome, all cases of which were grade 1 or 2. Two patients (5%) had grade 1 or 3 immune effector cell-associated neurotoxicity syndrome. These findings support that anti-BCMA/GPRC5D bispecific CAR T cells induced a high

response rate in patients with RRMM with extraosseous EMD, and the safety profile was manageable. This ongoing trial is registered at www.clinicaltrials.gov as NCT05509530.

Introduction

Extramedullary multiple myeloma (MM), characterized by infiltrated clonal plasma cells in tissues outside the bone marrow, is a highly aggressive subtype of MM.¹⁻³ The first-line treatments of MM, including immunomodulatory drugs, proteasome inhibitors, and monoclonal antibodies, have shown limited efficacy in patients with extramedullary disease (EMD).^{4,5} Chimeric antigen receptor (CAR) T cells targeting B-cell maturation antigen (BCMA) or G protein-coupled receptor, class C group 5 member D (GPRC5D) have yielded remarkable efficacy in relapsed or refractory MM (RRMM),⁶⁻⁹ but the response rate and remission depth seem less than satisfactory in extraosseous EMD.¹⁰⁻¹³ There is an urgent need for optimal therapies in patients with MM with EMD, especially

those with extraosseous EMD, which is more aggressive than paraosseous EMD.¹³

EMD subclones are highly heterogeneous, which could contain clones with negative mutations of BCMA.^{14,15} CAR T-cell therapy targeting dual/multiple antigens shows a potential to overcome antigen-negative escape and improve the effectiveness.¹⁶⁻¹⁸ A preclinical study has demonstrated CAR T cells targeting BCMA and GPRC5D could reduce relapse due to BCMA antigen escape and eliminate BCMA-negative MM cells in mouse models.¹⁹ Kumar et al reported that dual targeting of BCMA and GPRC5D with talquetamab and teclistamab enhanced efficacy for EMD in patients with RRMM, with an overall response rate (ORR) of 79% compared with single targeting monotherapy.^{20,21} In our previous study, we developed

anti-BCMA/GPRC5D bispecific CAR T cells and reported manageable safety and encouraging activity in patients with RRMM in a phase 1 study.²² These studies demonstrated that dual targeting of BCMA and GPRC5D exhibited advantages and had potential to become a more promising treatment option for RRMM, especially for RRMM with EMD, considering the highly heterogeneous BCMA and GPRC5D expression in plasmacytomas. In addition, single-target CAR T-cell therapy targeting BCMA or GPRC5D appeared to show superior early responses compared with their bispecific antibody counterparts.^{6,7,9,21} Based on these findings, we investigated the activity and safety of bispecific CAR T cells in patients with RRMM with extraosseous EMD in this phase 2 study.

Methods

Study design and participants

This open-label, single-arm, phase 2 study was conducted from 11 April 2023 to 21 July 2025 at The Affiliated Hospital of Xuzhou Medical University (Xuzhou, China). The study was reviewed and approved by the ethics committee of The Affiliated Hospital of Xuzhou Medical University. Written informed consent was obtained from each patient before study entry, in accordance with the Declaration of Helsinki. This study is registered at [ClinicalTrials.gov](https://clinicaltrials.gov) (identifier: NCT05509530).

Patients were eligible if they were aged 18 to 75 years, had histologically confirmed MM, and met the International Myeloma Working Group (IMWG)²³ diagnostic criteria for RRMM. EMD was defined as bone-independent tumors of plasma cells growing at anatomical sites outside of the bone marrow and were nonbone contiguous. EMD was confirmed by whole-body positron emission tomography-computed tomography (PET-CT) scans, as well as magnetic resonance imaging (MRI); CT was permitted as an alternative to PET-CT and MRI. Bone-dependent and paramedullary disease alone were not considered as EMD. All enrolled patients had an Eastern Cooperative Oncology Group score ≤ 3 , life expectancy ≥ 12 weeks, and adequate organ function (alanine aminotransferase < 3 times the upper limit of normal [ULN], aspartate aminotransferase < 3 times ULN, and total bilirubin < 3 times ULN). Patients could not be pregnant or lactating and had no plans for pregnancy within 6 months of treatment, due to concern for fetal toxicity of conditioning chemotherapy and unknown effects of CAR T cells. Patients were excluded if they had any mental or psychological illnesses that could potentially interfere with the completion of treatment; uncontrolled dysfunction of important organs; any active infections with HIV, *Mycobacterium tuberculosis*, hepatitis B virus, or hepatitis C virus; and highly allergic constitution or history of severe allergies, especially to interleukin-2. Other additional details of inclusion and exclusion criteria are provided in the protocol.

Procedures

Peripheral lymphocytes were isolated from patients by apheresis, and CD3⁺ T cells were sorted and stimulated with human T-activator CD3/CD28 dynabeads (Thermo Fisher Scientific, Waltham, MA) for CAR T-cell manufacture. A lentiviral vector was used to carry a bispecific CAR targeting BCMA and GPRC5D. The antigen recognition domain of the vector was constructed in a novel configuration with a common 4-1BB

costimulatory domain and CD3 ζ signaling domains. The recognition domain is composed of 2 single-chain fragment variable domains (scFvs) for BCMA and GPRC5D, which were derived from a murine hybridoma cell line and a human phage display antibody library, respectively (supplemental Figure 1, available on the *Blood* website). The manufacture of CAR T cells was commenced on the day of apheresis and completed in 7 to 14 days. The transduction efficiency and viability were examined by flow cytometry after transduction and confirmed on the day of CAR T-cell infusion.

Quality control and safety of the CAR T-cell product is indeed very important for this clinical trial. We did the following to ensure the consistency and safety of the CAR T-cell product: all of the lentiviral vector production and CAR T-cell preparation procedures were performed in the facility of Shanghai YaKe Biotechnology Ltd, which is certified by Shanghai Center for Disease Control and Prevention and monitored daily to guarantee that it was free from bacterial, fungal, and mycoplasma contamination. Preclinical evaluation of replication-competent lentivirus was performed to guarantee the safety of lentiviral vector products. Transduction efficiency was evaluated at least twice during CAR T-cell preparation: once at 48 hours after transduction and again on the last day of cell preparation, by flow cytometry for CAR expression using biotinylated goat anti-mouse immunoglobulin G (catalog no. 115-066-006; Jackson ImmunoResearch), followed by fluorescein isothiocyanate (FITC)-conjugated streptavidin (catalog no. 554067; Becton, Dickinson and Company). The testing for bacterial, fungal, and mycoplasma contamination was performed throughout CAR T-cell preparation, as well as on the day of final release of the cell product.

Anti-BCMA/GPRC5D bispecific CAR T cells were administered at a dose of 2.0×10^6 CAR-positive T cells per kg according to our previous phase 1 dose-escalation study.²² All patients needed to have measurable disease before infusion. Conditioning chemotherapy consisted of fludarabine 30 mg/m² (days -5 to -3) daily and cyclophosphamide 750 mg/m² (days -5) daily (supplemental Figure 2). Bridging therapy was allowed between apheresis and conditioning chemotherapy.

Activity was evaluated on days 14 and 28 (± 1 day) and at each follow-up session (2, 3, 6, and 12 months) after CAR T-cell infusion. Extramedullary myeloma response was assessed at the time of suspected first response and every 4 weeks from month 1 to month 3 and every 12 weeks from month 4 to disease progression. PET/CT was recommended when all plasmacytomas completely resolved in MRI or CT. Safety was assessed during the first month after CAR T-cell infusion, and all adverse events were fully documented. Follow-up assessments for safety were done every month from month 1 to month 6 and every 3 months from month 6 to month 12. After 1 year, follow-up assessments for activity and safety were continued. The amplification of anti-BCMA/GPRC5D CAR T cells in patients' peripheral blood after infusion was assessed by flow cytometry (supplemental Methods).

Outcomes

The primary end point was activity, including clinical response and measurable residual disease (MRD) in the bone marrow. Overall response included stringent complete response (sCR),

complete response (CR), very good partial response (VGPR), and partial response (PR), which was evaluated according to the IMWG consensus.²⁴ EMD responses were evaluated by decrease in size of the plasmacytomas by imaging examinations including PET/CT, MRI, and CT. Because the IMWG criteria do not include specific response criteria for imaging examinations, adaptations were incorporated to assess extramedullary myeloma response. The response criteria regarding EMD required the disappearance of all plasmacytomas for CR and sCR, a decrease $\geq 90\%$ in the sum of the maximal perpendicular diameter for a VGPR, a decrease $\geq 50\%$ for a PR, 25% to 49% decrease for a minimal response (MR), the recurrence of a known plasmacytoma that had disappeared with therapy or the appearance of any new area of extramedullary involvement or the increase $\geq 25\%$ in sum of the maximal perpendicular diameter of existing lesions for a progressive disease (PD), and case of not meeting the criteria for CR, VGPR, PR, MR, or PD for stable disease.^{10,20} Other required intramedullary and serological response criteria for CR, VGPR, PR, MR, PD, and stable disease were per IMWG response criteria.²⁴ ORR was defined as the proportion of patients obtaining a PR or better. MRD was monitored using flow cytometry, and MRD negativity was defined as the absence of phenotypically aberrant clonal plasma cells with a sensitivity of 10^{-5} .²⁵ progression-free survival (PFS) and overall survival (OS) were also evaluated in our study.

Safety was evaluated as the secondary end point of this study. The severity of adverse events was graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. Early and late hematologic toxicity were also evaluated with the European Society for Blood and Marrow Transplantation and the European Hematology Association consensus of immune effector cell-associated hematotoxicity (ICAHT) criteria.²⁶ Cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS) were defined and graded according to the American Society for Transplantation and Cellular Therapy criteria.²⁷ Local CRS was defined as localized inflammatory reactions associated with CAR T-cell treatment and manifested as redness, local pain, and inflammatory edema.^{10,28} B-cell aplasia was defined as $<1\%$ CD19⁺ peripheral blood mononuclear cells. Anti-BCMA/GPRC5D CAR T cells were detected after infusion as planned in the protocol to assess their amplification and persistence in vivo.

Statistics

This is an exploratory study, and all the analyses are descriptive in nature. The planned sample size was based on clinical and practical considerations to enable exploratory evaluations of activity and safety. As prespecified in the protocol, activity and safety analyses were done in all patients who received the anti-BCMA/GPRC5D bispecific CAR T cells. The all-treated population was the same as the intention-to-treat population. The cutoff date for all analyses was 2 November 2025. Statistical analysis was performed using SPSS 18.0 software for Windows (SPSS, Chicago, IL) and GraphPad Prism 6.0 (GraphPad Software, La Jolla, CA). Data are expressed as median and interquartile range (IQR) for continuous variables and frequency for categorical variables. Comparisons between groups were tested by the Mann-Whitney *U* test for 2 groups and the Kruskal-Wallis test for multiple groups. The Kaplan-Meier

method was used to evaluate PFS and OS. PFS was calculated from the date of CAR T-cell infusion to either the date of progression, the date of death from any cause, or the date of last contact (censored). OS was calculated from the date of CAR T-cell infusion to either the date of death from any cause or the date of last contact (censored). A *P* value $< .05$ (2-tailed) was considered statistically significant.

Results

Baseline characteristics of patients

Between 11 April 2023 and 21 July 2025, we screened 40 patients with RRMM with extraosseous EMD. Three patients were excluded because they did not meet the eligibility criteria. Thirty-seven patients were enrolled and underwent apheresis. All 37 patients received CAR T-cell infusion and were evaluated for activity and safety (Figure 1). Baseline characteristics are shown in Table 1 and supplemental Tables 1-3. Before lymphodepletion, 31 patients (84%) had varying degrees of cytopenia, including 10 (27%) with neutropenia, 29 (78%) with anemia, and 14 (38%) with thrombocytopenia. The percentage of infused cells expressing the anti-BCMA/GPRC5D bispecific CAR was determined on the day of infusion (supplemental Table 4). Before receiving CAR T cells, myeloma cells were detectable by flow cytometry in the bone marrow of 28 patients, among whom 28 were BCMA positive and 25 were GPRC5D positive. Fourteen patients (38%) received bridging therapies after apheresis, and the responses to bridging therapies before CAR T-cell infusion are provided in supplemental Table 5. In the 37 enrolled patients, 3 (8%) had previously received BCMA- or GPRC5D-targeted CAR T-cell therapies.

Activity

All 37 patients receiving CAR T-cell infusion were assessed for clinical responses (Figure 2) during a median follow-up of

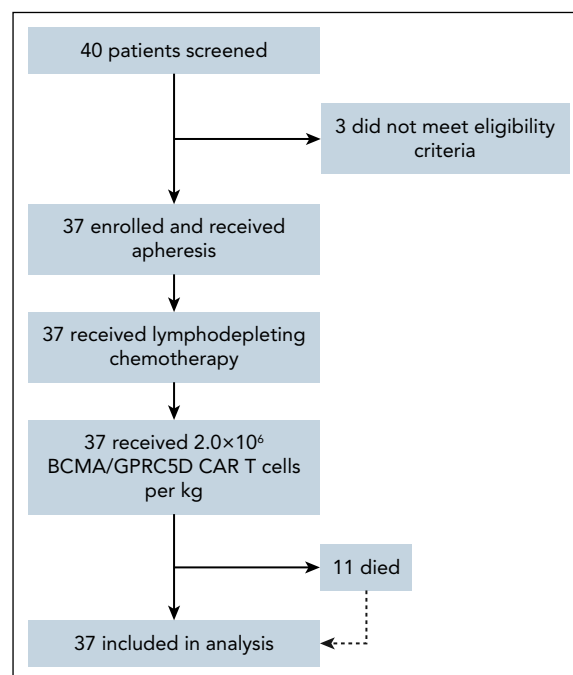


Figure 1. Flow diagram.

Table 1. Baseline characteristics of patients

	Patients who received CAR T cells (n = 37)
Median age, y	56 (53-61)
Sex	
Male	20 (54)
Female	17 (46)
Race	
Chinese	37 (100)
Other	0
ECOG performance status score*	
0	2 (5)
1	6 (16)
2	18 (49)
3	11 (30)
Median time from diagnosis, mo	39 (18-72)
Monoclonal globulin	
IgG	17 (46)
IgA	11 (30)
IgD	1 (3)
Light chain	6 (16)
Nonsecretory	2 (5)
High-risk cytogenetic profile†	17 (46)
R-ISS	
I	3 (8)
II	19 (51)
III	15 (41)
Median lines of previous therapies	5 (3-6)
Autologous hematopoietic stem cell transplantation	14 (38)
Previous proteasome inhibitors	
Bortezomib	
Exposed	37 (100)
Refractory‡	29 (78)
Ixazomib	
Exposed	15 (41)
Refractory	13 (35)
Carfilzomib	
Exposed	28 (76)
Refractory	22 (59)
Previous immunomodulatory drugs	
Thalidomide	
Exposed	13 (35)
Refractory	10 (27)

Table 1 (continued)

	Patients who received CAR T cells (n = 37)
Lenalidomide	
Exposed	34 (92)
Refractory	30 (88)
Pomalidomide	
Exposed	24 (65)
Refractory	21 (57)
Previous anti-CD38 monoclonal antibodies	
Exposed	31 (84)
Refractory	26 (70)
Previous selinexor	
Exposed	14 (38)
Refractory	10 (27)
Double-class refractory disease§	33 (89)
Triple-class refractory disease	25 (68)
Pentarefractory disease¶	11 (30)

Data are presented as median (IQR) or n (%). Percentages may not total 100 because of rounding.

ECOG, Eastern Cooperative Oncology Group; IgG, immunoglobulin G; R-ISS, Revised International Staging System.

*ECOG performance status scores range from 0 to 5, with higher scores indicating greater disability.

†High-risk cytogenetic profile was defined as the presence of del(17p), t(4;14), t(14;16), and amp(1q).

‡Refractory was defined as disease progression on or within 60 days after the last dose of the most recent drug given in each drug class.

§Double-class refractory disease was defined as disease refractory to an immunomodulatory agent and a proteasome inhibitor.

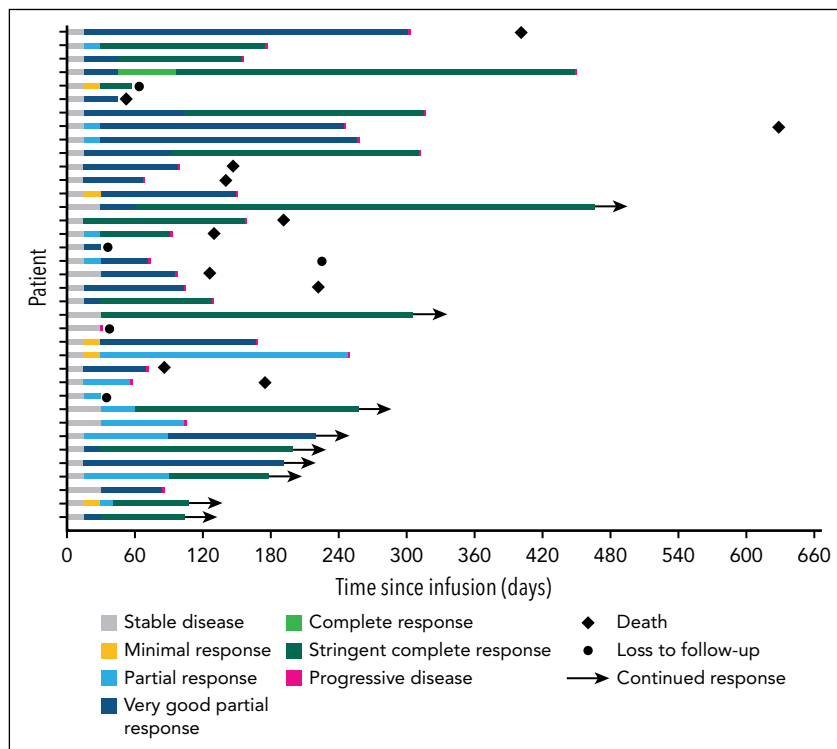
||Triple-class refractory disease was defined as disease refractory to an immunomodulatory agent, a proteasome inhibitor, and an anti-CD38 monoclonal antibody.

¶Pentarefractory disease was defined as disease refractory to lenalidomide, pomalidomide, bortezomib, carfilzomib, and an anti-CD38 monoclonal antibody.

10.1 months (IQR, 6.4-19.1) before the cutoff date (2 November 2025). The ORR was 97% (36/37) in the entire cohort, including 16 (43%) sCRs, 16 (43%) VGPRs, and 4 (11%) PRs. The median time to best response was 30.0 days (IQR, 15.0-32.8) after CAR T-cell infusion. Among the 36 patients with negative MRD detection in the bone marrow after CAR T-cell infusion, 35 (95%) were negative by day 28, and 1 (3%) was negative at 4 months. EMD completely resolved in 16 patients (43%) and partially shrunk in 20 patients (54%) evaluated by imaging examinations.

During follow-up, 24 patients (65%) progressed before the cutoff date. Among these patients, 9 manifested as increased serum M protein (including increased difference between involved and uninvolved serum free light-chain levels >10 mg/dL), 7 manifested as new or increased size of soft tissue plasmacytomas, and 8 patients showed combined manifestations of the above. Nine patients (24%) remained in remission,

Figure 2. Clinical responses of anti-BCMA/GPRC5D bispecific CAR T cells.



including 7 (19%) with sCR and 2 (5%) with VGPR. At progression, bone marrow MRD was detectable in 7 patients (19%) whose BCMA and GPRC5D expression were all positive. Bone marrow MRD remained negative in the other 30 patients (81%) during follow-up (Figure 2). The median duration of response was 5.0 months (95% confidence interval [CI], 1.8-8.3). The median PFS was 5.8 months (95% CI, 2.2-9.4), the median OS was not reached, and the estimated rate of OS was 70% (95% CI, 50-83) at 12 months and 50% (95% CI, 22-73) at 24 months (Figure 3). The PFS in this study is shorter than the PFS (25.5 months [95% CI, 13.8-37.1]; $P < .0001$) of patients without EMD from our previous phase 1 trial after treatment with anti-BCMA/GPRC5D bispecific CAR T cells (supplemental Figure 3).

Safety

The most common adverse events were hematologic toxicities (Table 2). A total of 37 patients (100%) had grade 3 to 4 hematologic toxicities (except lymphopenia), including 31 (84%) with neutropenia, 23 (62%) with anemia, and 16 (43%) with thrombocytopenia. The median durations of grade 3 to 4 neutropenia, anemia, and thrombocytopenia were 4.0 days (IQR, 2.5-6.5), 14.0 days (IQR, 7.0-18.0), and 19.0 days (IQR, 14.5-22.0), respectively. Patients with EMD appeared to have more severe early ICAHT ($P = .045$) than those without EMD from our previous phase 1 study (supplemental Figure 4).

CRS occurred in 27 patients (73%), with a median onset of 7.0 days (IQR, 4.5-8.0) after CAR T-cell infusion and a median duration of 4.0 days (IQR, 2.0-4.0). The median time from CAR T-cell infusion to resolution of CRS was 10.0 days (IQR, 8.5-12.0). All cases of CRS in these patients were grade 1 or 2 (Table 2). Local CRS occurred in 4 patients (11%). Sixteen patients received dexamethasone, 5 patients received dexamethasone

plus tocilizumab to control CRS, and all cases of CRS were rapidly relieved. The median times for C-reactive protein, interleukin-6, and ferritin to reach their peak concentrations in patients with CRS were 9.0 days (IQR, 8.0-13.0), 11.0 days (IQR, 6.5-16.0), and 12.0 days (IQR, 11.0-18.0) after infusion, respectively (supplemental Figure 5). It seemed that higher grade of CRS was associated with longer duration of hematologic toxicities (supplemental Figure 6), and longer duration of CRS was related to longer duration of grade 3 to 4 anemia (supplemental Figure 7). No significant differences were observed between clinical responses and severity of CRS (supplemental Figure 8), but better responses seemed to be associated with shorter duration of CRS in this cohort (supplemental Figure 9).

ICANS occurred in 2 patients (5%), of whom 1 patient had grade 1 ICANS, which occurred on day 6 after infusion and lasted for 2 days, and the other patient had grade 3 ICANS, which occurred on day 35 after infusion and lasted for 10 days. The patient with grade 3 ICANS experienced a sudden epileptic seizure (grade 3) with loss of consciousness, trismus, and limb convulsions. MRI imaging showed no significant abnormalities, and no GPRC5D-associated cerebellar dysfunctions were observed. The symptoms of both patients resolved after treatment with dexamethasone.

On-target, off-tumor toxicities related to GPRC5D expression in the skin and keratinized tissue manifested as grade 1 nail loss in 7 patients (19%), grade 1 pruritus in 4 (11%), rash in 3 (8%), and dry skin in 2 (2%; Table 2). Nail disorders resolved without any treatment. Rash, dry skin, and pruritus were treated with topical intervention and disappeared rapidly. No therapy-related cerebellar toxicities were observed by the cutoff date in our study.

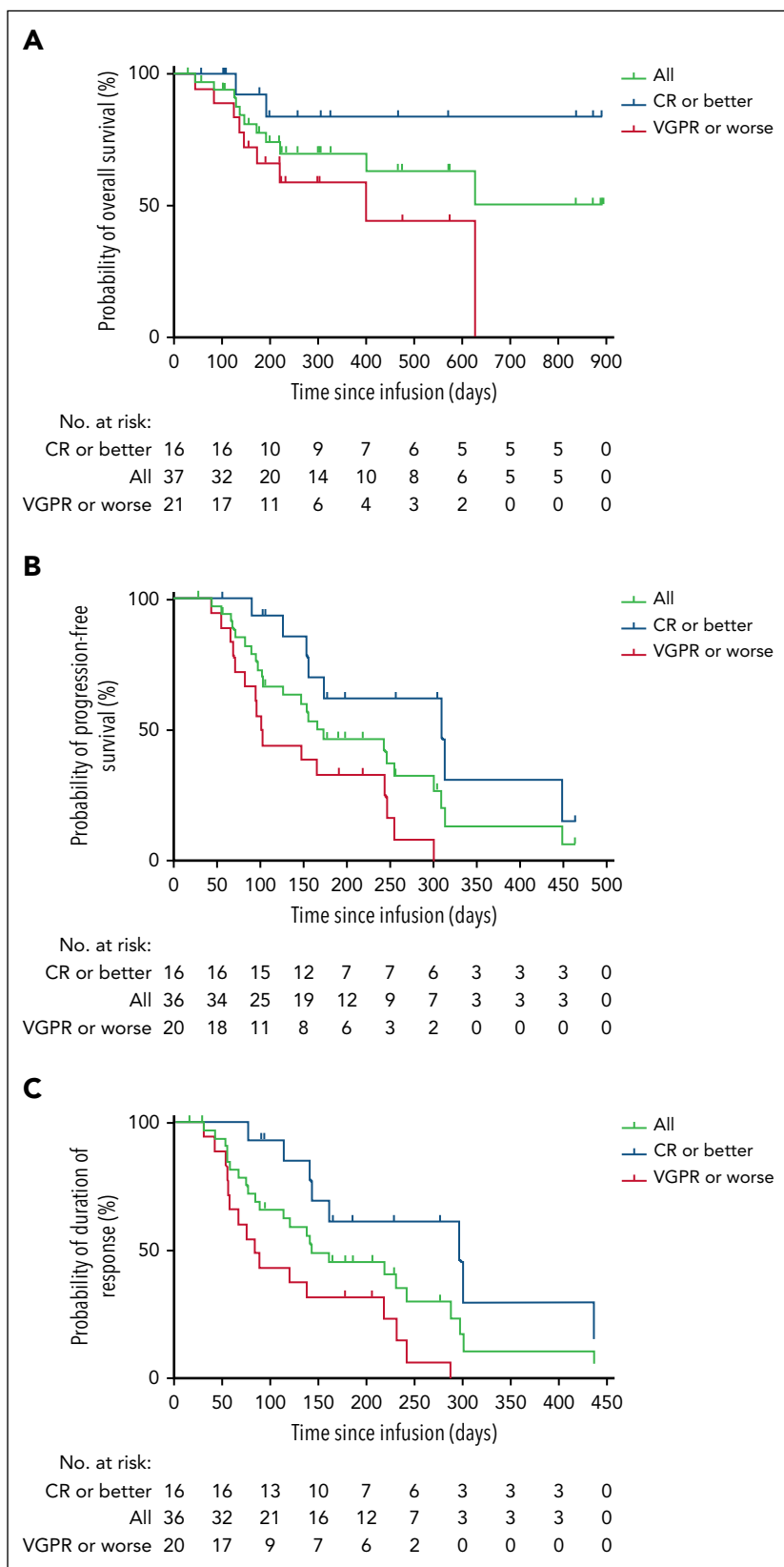


Figure 3. Kaplan-Meier analysis of DOR, PFS, and OS. (A) Kaplan-Meier curves of OS. (B) Kaplan-Meier curves of PFS. (C) Kaplan-Meier curves of DOR. Tick marks indicate the time of data censoring at the last follow-up. DOR, duration of response.

Twenty-seven patients (93%) developed B-cell aplasia after CAR T-cell infusion among 29 evaluable patients. B-cell aplasia recovered in 13 patients (45%) within 3 months after CAR T-cell infusion.

Severe hypogammaglobulinemia (immunoglobulin G <400 mg/dL) was observed in 19 patients (51%), and 11 patients (30%) recovered within 3 months after CAR T-cell infusion.

Table 2. Adverse events

Adverse event	Grade 1-2	Grade 3	Grade 4
Hematologic			
Leukopenia	6 (16)	13 (35)	18 (49)
Neutropenia	6 (16)	15 (41)	16 (43)
Lymphopenia	2 (5)	6 (16)	29 (78)
Anemia	12 (32)	13 (35)	10 (27)
Thrombocytopenia	11 (30)	5 (14)	11 (30)
Gastrointestinal disorders			
Nausea	12 (32)	0	0
Vomiting	3 (8)	0	0
Diarrhea	7 (19)	0	0
Abdominal distension	2 (5)	0	0
Metabolism and nutrition disorders			
Hypokalemia	18 (56)	3 (8)	0
Hyponatremia	13 (35)	3 (8)	0
Hypocalcemia	24 (65)	3 (8)	0
Hypoalbuminemia	16 (43)	0	0
Respiratory and cardiac disorders			
Productive cough	6 (16)	0	0
Dyspnea	4 (11)	0	0
Sinus tachycardia	5 (14)	0	0
Diseases of the skin and related tissues			
Pruritus	4 (11)	0	0
Rash	3 (8)	0	0
Dry skin	2 (5)	0	0
Nail changes	7 (19)	0	0
Neurotoxicities			
ICANS*	1 (3)	1 (3)	0
Headache	5 (29)	0	0
Others			
Decreased fibrinogen	6 (16)	5 (14)	0
Prolonged APTT	12 (32)	0	0
Increased ALT	6 (16)	0	0
Increased AST	12 (32)	0	0
Increased GGT	9 (24)	0	0
Increased bilirubin	2 (5)	0	0
Fatigue	7 (19)	0	0
Insomnia	2 (5)	0	0
CRS*	27 (73)	0	0
IEC-HS*	0	1 (3)	0

Adverse events were graded according to National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. Data are presented as n (%). Percentages may not total 100 because of rounding.

APTT, activated partial thromboplastin time; ALT, alanine aminotransferase; AST, aspartate aminotransferase; GGT, γ -glutamyltransferase.

*ICANS, CRS, and IEC-HS were graded according to the American Society for Transplantation and Cellular Therapy criteria.

Emerging infections were observed in 12 patients (32%), including 6 patients with viral infections, 4 with bacterial infections, and 5 with fungal infections. Viral infection was the

most common infection after CAR T-cell infusion (supplemental Table 6). All 12 patients had emerging active infections within 3 months, and 1 patient had infections after 3 months after

CAR T-cell infusion. The most common site of infections was the lung (5 [14%]), followed by the skin (4 [11%]), upper respiratory tract (3 [8%]), and urinary tract (3 [8%]).

Other grade 3 or worse nonhematologic toxicities included immune effector cell–associated hemophagocytic lymphohistiocytosis–like syndrome (IEC-HS), electrolyte disturbance, and decreased fibrinogen. One patient had grade 3 IEC-HS on day 32 after infusion. The symptoms resolved after treatment with dexamethasone and other supportive care. Electrolyte disturbance and decreased fibrinogen resolved rapidly with supportive treatments (Table 2). Eleven patients (30%) died of disease progression during follow-up, and there were no treatment-related deaths in the trial.

CAR T-cell expansion and persistence

The presence of anti-BCMA/GPRC5D CAR T cells in patients' peripheral blood after infusion was detected by flow cytometry (supplemental Figure 10). In patients with assessable CAR T-cell amplification, anti-BCMA/GPRC5D peaked between 2 and 4 weeks in most patients after infusion, followed by a gradual decrease. The rates of detectable CAR T cells in peripheral blood at 1 and 3 months were 85% and 53%, respectively. Eleven patients (30%) had detectable CAR T cells for >5 months. No significant differences were observed for the peak levels of CAR expansion and the CAR expansion at 3 months between patients with EMD and those without EMD from our previous phase 1 trial (supplemental Figure 11). No significant differences were observed between peak levels of anti-BCMA/GPRC5D CAR T-cell expansion and clinical responses (supplemental Figure 12).

Discussion

Although new drugs have improved the prognosis of MM, EMD remains a clinical challenge, with poor responses in relapsed or refractory patients.¹⁻⁴ Although CAR T-cell therapy has revolutionized immunotherapy for RRMM, outcomes for patients with extraosseous EMD still remain unsatisfactory, characterized by low response rates and poor long-term survival, even with BCMA-targeted treatments.^{10-13,29,30} To address this unmet need, we developed anti-BCMA/GPRC5D bispecific CAR T cells to improve outcomes in this high-risk population.

When malignant plasma cells invade extramedullary sites, they resemble solid tumors, exhibiting antigen heterogeneity and a complex microenvironment in the solid extramedullary mass.^{2,4} Substantial research has targeted overcoming CAR T-cell therapy limitations in solid tumors, in which immune escape, primarily from insufficient antigen expression and tumor heterogeneity, remains a critical barrier.^{31,32} Dual-targeting strategies have thus emerged to broaden antigen coverage and overcome immune escape.¹⁶⁻¹⁸ In addition, BCMA-negative disease was rarely reported in patients with newly diagnosed MM, but monoallelic loss of BCMA was observed in 8.58% of the newly diagnosed patients, which was considered as a high-risk factor for BCMA antigen loss.^{33,34} BCMA antigen loss occurred primarily in relapsed patients and was reported in ~4% to 6% of relapsed patients who were treated with idecabtagene vicleucel.^{6,35} The structure of anti-BCMA/GPRC5D

CAR T cells is engineered with anti-BCMA and anti-GPRC5D scFvs in the same CAR at a 1:1 ratio to increase tumor antigen coverage. Both scFvs can bind to myeloma antigens to mitigate relapse or treatment failure of single-target CAR T cells. This bispecific structure of anti-BCMA/GPRC5D CAR T cells ensures that CAR T cells can fully recognize and bind to tumor antigens, thereby reducing off-target effects and enhancing their specificity.

In this phase 2 trial, anti-BCMA/GPRC5D bispecific CAR T cells demonstrated promising efficacy in heavily pretreated, patients with RRMM with extraosseous EMD. The ORR was 97% (36/37), with all responders achieving MRD negativity in the bone marrow. Most patients had high-risk features, including advanced stage, high-risk cytogenetics, or disease progression after prior CAR T-cell therapy.

Compared with existing therapies, including proteasome inhibitors, immunomodulatory drugs, monoclonal antibodies, and BCMA-targeted treatments, anti-BCMA/GPRC5D CAR T cells yielded higher response rates (ORR, 97%) and deeper responses (43% sCR), with encouraging PFS and OS in relapsed or refractory patients with EMD. The ORR in patients with secondary EMD receiving novel agents was ≤50% in most reports. Recent studies demonstrated dismal outcomes, with a median PFS of 2.2 months for proteasome inhibitors and immunomodulatory drug-based combinations and 2.9 months for alkylator-based combinations.³⁰ Teclistamab led to worse ORR in patients with EMD (36% vs 69%) than those without it in the pivotal phase 2 MajesTec-1 study, and the data of teclistamab use in the real-world setting demonstrate a dismal median PFS of 2.1 months in patients with EMD.³⁶ The results of our study indicated that dual-targeting strategies with anti-BCMA/GPRC5D CAR T cells may enhance treatment potency, maximizing tumor eradication in heterogeneous cell populations in patients with EMD. This may be explained by broader antigen coverage, which helps overcome heterogeneity, prevents antigen escape, and eradicates divergent myeloma clones.

In a recent multicenter retrospective analysis of patients with EMD who were treated with commercial BCMA CAR T-cell therapy, 58% of patients with EMD obtained a response, and 28% of patients achieved a CR or better, with a median PFS of 5.0 months and a median OS of 12.2 months,¹² which is consistent with previously reported data from our center.¹⁰ Similarly, Zanwar et al reported that the median PFS was 5.3 months (95% CI, 4.1-6.9) for EMD patients with idecabtagene vicleucel.¹¹ Although Martin et al reported more promising results that patients with relapsed/refractory myeloma with baseline plasmacytomas achieved a PFS for 13.8 months (95% CI, 5.3 to not estimable) after treatment with ciltacabtagene autoleucel in CARTITUDE-1 study.⁷ However, the enrolled patients with plasmacytomas included paraneoplastic EMD. Extraosseous EMD but not paraneoplastic diseases was thought to portend inferior outcomes after CAR T-cell therapy in MM.¹³ In RedirecTT-1 study, dual targeting with talquetamab plus teclistamab achieved an ORR of 79%, with a median duration of response of 13.8 months (95% CI, 11.5 to not estimable) in patients with extraosseous EMD.²⁰ Anti-BCMA/GPRC5D bispecific CAR T cells exhibited superior early response rate compared to talquetamab plus teclistamab, but the PFS

seemed less satisfactory, which might be due to the readily available off-the-shelf advantages of bispecific antibodies and being suitable for repeated administration to extend remission period in patients with EMD.

The response rate was very promising with an ORR of 97% in this study, but the median PFS was modest, which was roughly in line with the PFS for BCMA-targeted therapies.¹⁰⁻¹² The data in this study demonstrated that dual targeting to increase antigen coverage had potential to overcome heterogeneous BCMA and GPRC5D expression and improve early responses. However, we and others previously showed the solid tumor-like microenvironment of EMD is quite different from the bone marrow,^{10,14,15} which might inhibit the persistence of CAR T cells in plasmacytomas and impair the long-term efficacy. This is still a great challenge for patients with RRMM with EMD and deserved further explorations in the future.

Patients with EMD appeared to have more severe early ICAHT than patients without EMD, which was consistent with previous study,³⁷ and 6 of 37 patients (16%) had late ICAHT after anti-BCMA/GPRC5D CAR T-cell infusion. All 6 patients had varying degrees of cytopenia at baseline before lymphodepletion. Close attention should be paid to prolonged hematologic toxicities, for which the underlying mechanisms are still unclear, because they were associated with life-threatening complications and poor outcomes after CAR T-cell therapy in patients with MM.³⁸ Compared to monotherapy or combination therapy with bispecific antibodies, the safety profiles were similar for the incidence and severity of adverse events.^{20,21,36} Additionally, Hematologic toxicities, CRS (70%; all grade 1-2), and neurotoxicity (5%) were also comparable to single-target CAR T-cell products, with no increase in severity.⁶⁻⁸ Grade 3 IEC-HS was observed in 1 patient (3%). The symptoms of CRS, ICANS, and IEC-HS were resolved with glucocorticoids or tocilizumab (or both), and other severe adverse events resolved rapidly with supportive care, demonstrating the manageable safety of anti-BCMA/GPRC5D CAR T cells.

Some limitations of this study should be considered. First, although the pretreatment profile of patients in this study was similar to that of patients from other CAR T-cell trials in China, ~16% of patients had no prior CD38 monoclonal antibody exposure, and 38% of patients had prior autologous stem cell transplantation, which made the cohort seem less heavily pretreated than some of the CAR T-cell studies in the United States,⁶⁻⁸ and the differences might have a favorable effect on the observed activity of bispecific CAR T cells. Second, the patients enrolled in this study were from single center in China, and the trial was designed as a single-arm, nonrandomized study, which might increase research bias. In future studies, we plan to enlarge sample size and extend the follow-up period to investigate the outcomes in these patients with anti-BCMA/GPRC5D CAR T-cell therapy.

In summary, anti-BCMA/GPRC5D bispecific CAR T cells show high efficacy and a manageable safety profile in patients with RRMM with extraosseous EMD. Further prospective studies with larger sample size and longer follow-up are warranted to confirm these encouraging results.

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Authorship

Contribution: D.Z., Y.Q., S.M., A.H.C., B.P., and Z.Y. were involved in study design, data analysis and interpretation, and manuscript writing; Q.S., W.G., J.X., X.Z., H.C., W.C., F.Z., K.Q., F.X., D.L., W.S., H.L., L.Z., H.Z., T.Q., and D.Y. enrolled and treated patients; A.H.C. contributed to vector design and developed the anti-B-cell maturation antigen/G protein-coupled receptor, class C group 5 member D chimeric antigen receptor (CAR) T cells; D.Z., Q.S., Y.Z., and S.P. contributed to the manufacturing and quality control of CAR T cells; D.Z., Y.Q., S.M., and L.Z. contributed to data collection and assembly; A.H.C., B.P., and Z.Y. had accessed and verified the data; and all authors had full access to all data in the study, approved the final version of the manuscript, and had the final responsibility for the decision to submit for publication.

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Footnotes

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The data sets supporting the findings of this study are included in the article and the supplemental Data. Additional deidentified individual patient data may be made available from corresponding author Zhiling Yan (hematology_md@126.com) upon reasonable request.

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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